

Teaching Notes on the Assignment Problem

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1. Introduction

The **Assignment Problem (AP)** is a special case of the transportation problem where the objective is to assign a set of *agents* (or workers, machines, etc.) to an equal number of *tasks* (or jobs) at a minimum total cost or maximum total profit.

Each agent can perform exactly one task, and each task must be performed by exactly one agent.

2. Mathematical Formulation

Suppose there are n agents and n tasks, and let

$$c_{ij} = \text{cost of assigning agent } i \text{ to task } j, \quad i, j = 1, 2, \dots, n.$$

Define a binary decision variable:

$$x_{ij} = \begin{cases} 1, & \text{if agent } i \text{ is assigned to task } j, \\ 0, & \text{otherwise.} \end{cases}$$

The **objective function** is:

$$\min Z = \sum_{i=1}^n \sum_{j=1}^n c_{ij} x_{ij}$$

subject to the constraints:

$$\begin{aligned} \sum_{j=1}^n x_{ij} &= 1, \quad i = 1, 2, \dots, n, && \text{(Each agent assigned one task)} \\ \sum_{i=1}^n x_{ij} &= 1, \quad j = 1, 2, \dots, n, && \text{(Each task assigned one agent)} \\ x_{ij} &\in \{0, 1\}, && \text{(Binary restriction)} \end{aligned}$$

3. Matrix Representation

The problem can be represented by a **cost matrix** $C = [c_{ij}]_{n \times n}$:

$$C = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n1} & c_{n2} & \cdots & c_{nn} \end{bmatrix}$$

Each row corresponds to an agent, and each column corresponds to a task.

4. Hungarian Method (Algorithmic Steps)

The **Hungarian Method** provides an efficient, step-by-step procedure to find the optimal assignment.

Step 1: Row Reduction

Subtract the smallest element in each row from all the elements of that row.

Step 2: Column Reduction

Subtract the smallest element in each column from all the elements of that column.

Step 3: Zero Allocation

In the reduced matrix (obtained from step 2 & 3), find assignment;

- Start examining rows; allocate the zero by a $\square$. If there is another zero in the corresponding column, cross them ($\times$) and move to next row. If a row have more than one zero then leave it temporarily.
- Once all rows are covered in the previous step; repeat the process for the columns.

If each row and column has a unique allocation then it is optimal; otherwise move to the next step.

Step 4: Covering Zeros

Cover all zeros in the resulting matrix using a minimum number of horizontal and vertical lines. To draw the minimum number of lines one can follow the process:

- Mark ($\checkmark$) rows that do not have any assigned zero ($\square$);
 - Mark ($\checkmark$) columns that have cross marked zero ($\times$) in the marked row;
 - Mark ($\checkmark$) rows that have assigned zero ($\square$) in the marked column;
 - Repeat the process until the chain is complete.
 - Draw lines through the unmarked row and marked columns. This will gives us desire minimum number of horizontal and vertical lines.
- If the minimum number of covering lines equals n , an optimal assignment exists among the zeros.
 - Otherwise, go to Step 4.

Step 5: Adjust the Matrix

Find the smallest uncovered element (say k). Subtract k from all uncovered elements and add k to all elements lying at the intersection of two covering lines. Return to Step 3.

Step 6: Optimal Assignment

Once there are n independent zeros, make an assignment such that:

- No two assigned zeros are in the same row or column.

The corresponding cells give the optimal assignment with minimum total cost.

5. Example

Problem: Use Hungarian method to assign four workers A_1, A_2, A_3, A_4, A_5 to four jobs J_1, J_2, J_3, J_4, J_5 to minimize total cost.

$$C = \begin{bmatrix} 11 & 7 & 10 & 17 & 10 \\ 13 & 21 & 7 & 11 & 13 \\ 13 & 13 & 15 & 13 & 14 \\ 18 & 10 & 13 & 16 & 14 \\ 12 & 8 & 16 & 19 & 10 \end{bmatrix}$$

Answer: Row reduction:

	J_1	J_2	J_3	J_4	J_5
A_1	4	0	3	10	3
A_2	6	14	0	4	6
A_3	0	0	2	0	1
A_4	8	0	3	6	4
A_5	4	0	8	11	2

Column reduction:

	J_1	J_2	J_3	J_4	J_5
A_1	4	0	3	10	2
A_2	6	14	0	4	5
A_3	0	0	2	0	0
A_4	8	0	3	6	3
A_5	4	0	8	11	1

Zero allocation:

	J_1	J_2	J_3	J_4	J_5
A_1	4	0	3	10	2
A_2	6	14	0	4	5
A_3	0	9	2	0	0
A_4	8	9	3	6	3
A_5	4	9	8	11	1

Covering the zeros:

	J_1	J_2	J_3	J_4	J_5
A_1	4	0	3	10	2 ✓
A_2	6	14	0	4	5
A_3	0	9	2	0	0
A_4	8	9	3	6	3 ✓
A_5	4	9	8	11	1 ✓

Adjust the matrix:

	J_1	J_2	J_3	J_4	J_5
A_1	3	0	2	9	1
A_2	6	15	0	4	5
A_3	0	1	2	0	0
A_4	7	0	2	5	2
A_5	3	0	7	10	0

Here the number of allocated cell is less than five, so need to repeat the process again.

	J_1	J_2	J_3	J_4	J_5
A_1	3	0	2	9	1
A_2	6	15	0	4	5
A_3	0	1	2	0	0
A_4	7	0	2	5	2
A_5	3	0	7	10	0

	J_1	J_2	J_3	J_4	J_5
A_1	2	0	1	8	0
A_2	6	16	0	4	5
A_3	0	2	2	0	0
A_4	6	0	1	4	1
A_5	3	1	7	10	0

	J_1	J_2	J_3	J_4	J_5
A_1	1	0	0	7	0
A_2	6	17	0	4	6
A_3	0	3	2	0	1
A_4	5	0	0	3	1
A_5	2	1	6	9	0

	J_1	J_2	J_3	J_4	J_5
A_1	0	0	0	6	0
A_2	5	17	0	3	6
A_3	0	4	3	0	2
A_4	4	0	0	2	1
A_5	1	1	6	8	0

This table have five allocation, so it achieved optimality. The optimal path is:

$$A_1 \rightarrow J_1, A_2 \rightarrow J_3, A_3 \rightarrow J_4, A_4 \rightarrow J_2 \text{ and } A_5 \rightarrow J_5.$$

Total Cost = $11 + 7 + 13 + 10 + 10 = 51$.

Problem: Use Hungarian method to assign four person A_1, A_2, A_3 to four jobs J_1, J_2, J_3, J_4 to minimize total cost under the restriction that the person P_3 can not do the job J_4 .

$$C = \begin{bmatrix} 20 & 25 & 22 & 28 \\ 15 & 18 & 23 & 27 \\ 19 & 17 & 21 & \infty \end{bmatrix}$$

Answer:

Since the given AP is unbalanced, we need to add a dummy person P_4 as the 4th row in the given matrix.

Addition of a dummy row

	J_1	J_2	J_3	J_4
A_1	20	25	22	28
A_2	15	18	23	17
A_3	19	17	21	∞
A_4	0	0	0	0

Row and Column reduction

	J_1	J_2	J_3	J_4
A_1	0	5	2	8
A_2	0	3	8	2
A_3	2	0	4	∞
A_4	0	0	0	0

Covering of zeros

	J_1	J_2	J_3	J_4
A_1	0	5	2	8 ✓
A_2	0	3	8	2 ✓
A_3	2	0	4	∞
A_4	0	0	0	0

Optimal Solution

	J_1	J_2	J_3	J_4
A_1	0	3	0	5
A_2	0	1	6	0
A_3	4	0	4	∞
A_4	2	0	0	0

Alternate Solution

	J_1	J_2	J_3	J_4
A_1	0	3	0	5
A_2	0	1	6	0
A_3	4	0	4	∞
A_4	2	0	0	0

Problem 1 Use Hungarian method to assign four person A_1, A_2, A_3 to four jobs J_1, J_2, J_3, J_4 to minimize total cost under the restriction that **the person P_3 assigned to the job J_4 .**

$$C = \begin{bmatrix} 20 & 25 & 22 & 28 \\ 15 & 18 & 23 & 27 \\ 19 & 17 & 21 & 11 \\ 10 & 15 & 7 & 12 \end{bmatrix}$$

Answer:

	J_1	J_2	J_3
A_1	20	25	22
A_2	15	18	23
A_4	10	15	7

Deletion of P_3 row and J_4 column

	J_1	J_2	J_3
A_1	0	5	2
A_2	0	3	8

Row reduction

A_4	3	8	0
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	J_1	J_2	J_3
A_1	0	2	2
A_2	0	0	8

Column reduction

A_4	3	5	0
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Therefore the optimal solution

$$A_1 \rightarrow J_1, A_2 \rightarrow J_2, A_3 \rightarrow J_4, A_4 \rightarrow J_3.$$

$$\text{Total Cost} = 20 + 18 + 11 + 12 = 56.$$

Problem 2 Consider the following **profit matrix**. Use Hungarian method to assign four person A_1, A_2, A_3 to four jobs J_1, J_2, J_3, J_4 to **maximize the profit**

$$C = \begin{bmatrix} 20 & 25 & 22 & 28 \\ 15 & 18 & 23 & 27 \\ 19 & 17 & 21 & 11 \\ 10 & 15 & 7 & 12 \end{bmatrix}$$

	Reduction of element from the maximum element				
	J_1	J_2	J_3	J_4	
A_1	8	3	6	0	
A_2	13	10	5	11	
A_3	9	11	7	17	
A_4	18	13	21	16	Row reduction
	J_1	J_2	J_3	J_4	
A_1	8	3	6	0	
A_2	8	5	0	6	
A_3	2	4	0	10	
A_4	5	0	8	3	Column reduction
	J_1	J_2	J_3	J_4	
A_1	6	3	6	0	
A_2	6	5	0	6	
A_3	0	4	0	10	
A_4	3	0	8	3	

Therefore the optimal solution

$$A_1 \rightarrow J_4, A_2 \rightarrow J_3, A_3 \rightarrow J_1, A_4 \rightarrow J_2.$$

$$\text{Total Cost} = 28 + 23 + 19 + 15 = 85.$$

Problem 3 Consider the following **profit matrix**. Use Hungarian method to assign four person A_1, A_2, A_3 to four jobs J_1, J_2, J_3, J_4 to **maximize the profit if the person P_1 has to do the job J_1**

$$C = \begin{bmatrix} 20 & 25 & 22 & 28 \\ 15 & 18 & 23 & 27 \\ 19 & 17 & 21 & 11 \\ 10 & 15 & 7 & 12 \end{bmatrix}$$

Answer: First remove the first row and first column from the given matrix. Then subtract all the elements from the maximum element of the reduced matrix, and then proceed similar as before (left as an exercise for the reader). The optimal solution will be

$$A_1 \rightarrow J_1, A_2 \rightarrow J_4, A_3 \rightarrow J_3, A_4 \rightarrow J_2.$$

$$\text{Total Cost} = 20 + 17 + 21 + 15 = 73.$$